[redacted]

8.21 A 60-turn coil carries a current of 2 A and lies in the plane $x + 2y - 5z = 12$ such that the magnetic moment **m** of the coil is directed away from the origin. Calculate **m**, assuming that the area of the coil is 8 cm^2.

Do any two for hw

Section 8.5—Magnetization in Materials

8.22 For a linear, isotropic, and homogeneous magnetic medium, show that $M = \dfrac{\chi_m}{\mu_o(1+\chi_m)} B.$

8.23 A block of iron ($\mu = 5000\mu_o$) is placed in a uniform magnetic field with 1.5 Wb/m^2. If iron consists of 8.5×10^{28} atoms/m^3, calculate (a) the magnetization **M**, (b) the average magnetic moment.

8.24 In a magnetic material, with $\chi_m = 6.5$, the magnetization is $\mathbf{M} = 24y^2\mathbf{a}_z$ A/m. Find μ_r, **H**, and **J** at $y = 2$ cm.

8.25 In a ferromagnetic material ($\mu = 4.5\mu_o$),

$$\mathbf{B} = 4y\mathbf{a}_z \text{ mWb/m}^3$$

calculate (a) χ_m, (b) **H**, (c) **M**, (d) $\mathbf{J}_b$.

8.26 An electromagnet is made of a ferromagnetic material whose magnetization curve can be approximated by

$$B(H) = B_oH/(H_o + H)\,\text{mWb/m}^3$$

where $B_o = 2$ Wb/m^2 and $H_o = 100$ A/m
Find μ_r when $H = 250$ A/m.

8.27 An infinitely long cylindrical conductor of radius a and permeability $\mu_o\mu_r$ is placed along the z-axis. If the conductor carries a uniformly distributed current I along $\mathbf{a}_z$, find **M** and $\mathbf{J}_b$ for $0 < \rho < a$.

Section 8.7—Magnetic Boundary Conditions

***8.28** (a) For the boundary between two magnetic media such as is shown in Figure 8.16, show that the boundary conditions on the magnetization vector are

$$\frac{M_{1t}}{\chi_{m1}} - \frac{M_{2t}}{\chi_{m2}} = K \quad \text{and} \quad \frac{\mu_1}{\chi_{m1}} M_{1n} = \frac{\mu_2}{\chi_{m2}} M_{2n}$$

(b) If the boundary is not current free, show that instead of eq. (8.49), we obtain

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}\left[1 + \frac{K\mu_2}{B_2\sin\theta_2}\right]$$

Do any two for hw

8.29 Region 1, for which $\mu_1 = 2.5\mu_o$, is defined by $z < 0$, while region 2, for which $\mu_2 = 4\mu_o$, is defined by $z > 0$. If $\mathbf{B}_1 = 6\mathbf{a}_x - 4.2\mathbf{a}_y + 1.8\mathbf{a}_z$ mWb/m^2, find $\mathbf{H}_2$ and the angle $\mathbf{H}_2$ makes with the interface.

8.30 If $\mu_1 = 2\mu_o$ for region 1 ($0 < \phi < \pi$) and $\mu_2 = 5\mu_o$ for region 2($\pi < \phi < 2\pi$) and $\mathbf{B}_2 = 10\mathbf{a}_\rho + 15\mathbf{a}_\phi - 20\mathbf{a}_z$ mWb/m^2. Calculate (a) $\mathbf{B}_1$, (b) the energy densities in the two media.

8.31 Region 1 ($x < 0$) is free space, while region 2 ($x > 0$) is a magnetic material with $\mu = 50\mu_o$. If $\mathbf{B}_1 = 40\mathbf{a}_x - 30\mathbf{a}_y + 10\mathbf{a}_z$ mWb/m^2, find $\mathbf{H}_2$.

8.32 Suppose space is divided into region 1 ($y < 0$, $\mu_1 = \mu_o\mu_{r1}$) and region 2 ($y < 0$, $\mu_2 = \mu_o\mu_{r2}$). If $\mathbf{H}_1 = \alpha\mathbf{a}_x - \beta\mathbf{a}_y + \delta\mathbf{a}_z$ A/m, find $\mathbf{H}_2$.

8.33 Region 1 is $y > 0$ with $\mu_1 = \mu_o$, while region 2 is $y < 0$ with $\mu_2 = 12\mu_o$. If $\mathbf{B}_2 = 1.4\mathbf{a}_x + 0.6\mathbf{a}_y - 2\mathbf{a}_z$ Wb/m^2, find $\mathbf{H}_1$ and $\mathbf{B}_1$.

***8.34** Region 1 is defined by $x - y + 2z > 5$ with $\mu_1 = 2\mu_o$, while region 2 is defined by $x - y + 2z < 5$ with $\mu_2 = 5\mu_o$. If $\mathbf{H}_1 = 40\mathbf{a}_x + 20\mathbf{a}_y - 30\mathbf{a}_z$ A/m, find (a) $\mathbf{H}_{1n}$, (b) $\mathbf{H}_{2t}$, (c) $\mathbf{B}_2$.

8.35 Inside a right circular cylinder, $\mu_1 = 800\,\mu_o$, while the exterior is free space. Given that $\mathbf{B}_1 = \mu_o(22\mathbf{a}_\rho + 45\mathbf{a}_\phi)$ Wb/m², determine $\mathbf{B}_2$ just outside the cylinder.

8.36 Medium 1 is free space and is defined by $r < a$, while medium 2 is a magnetic material with permeability μ_2 and defined by $r > a$. The magnetic flux densities in the media are:

$$\mathbf{B}_1 = B_{o1}\left[\left(1+\frac{1.6a^3}{r^3}\right)\cos\theta\,\mathbf{a}_r - \left(1+\frac{0.8a^3}{r^3}\right)\sin\theta\,\mathbf{a}_\theta\right]$$

$$\mathbf{B}_2 = B_{o2}(\cos\theta\, a_r - \sin\theta\,\mathbf{a}_\theta)$$

Find μ_2.

8.37 Region $0 \le z \le 2$ m is filled with an infinite slab of magnetic material ($\mu = 2.5\mu_o$). If the surfaces of the slab at $z = 0$ and $z = 2$, respectively, carry surface currents $30\mathbf{a}_x$ A/m and $-40\mathbf{a}_x$ as in Figure 8.44, calculate **H** and **B** for

(a) $z < 0$
(b) $0 < z < 2$
(c) $z > 2$

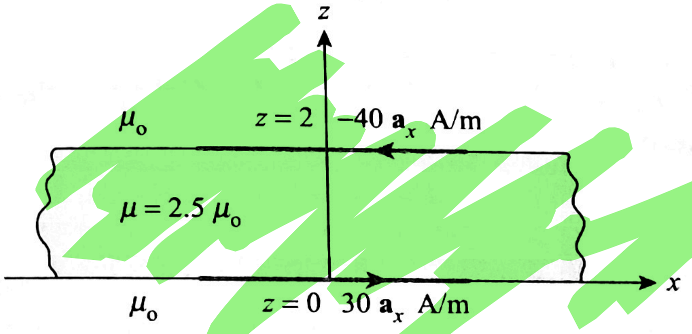

Figure 8.44 For Problem 8.37.

8.38 The plane $z = 0$ separates air ($z \ge 0, \mu = \mu_o$) from iron ($z \le 0, \mu = 200\mu_o$). Given that

$$\mathbf{H} = 10\mathbf{a}_x + 15\mathbf{a}_y - 3\mathbf{a}_z \text{ A/m}$$

in air, find **B** in iron and the angle it makes with the interface.

Section 8.8—Inductors and Inductance

***8.39** (a) If the cross section of the toroid of Figure 7.15 is a square of side a, show that the self-inductance of the toroid is

$$L = \frac{\mu_o N^2 a}{2\pi}\ln\left[\frac{2\rho_o + a}{2\rho_o - a}\right]$$

(b) If the toroid has a circular cross section as in Figure 7.15, show that

$$L = \frac{\mu_o N^2 a^2}{2\rho_o}$$

Do any one for hw

where $\rho_o \gg a$.

8.40 An air-filled toroid of square cross section has inner radius 3 cm, outer radius 5 cm, and height 2 cm. How many turns are required to produce an inductance of 45 μH?

8.41 Determine the inductance per unit length of a cylindrical conductor of radius 3 mm. Assume $\mu = \mu_o$.

8.42 A coaxial cable has an internal inductance that is twice the external inductance. If the inner radius is 6.5 mm, calculate the outer radius.

8.43 A coaxial cable has inner conductor of radius $a = 2.5$ mm and outer conductor of radius $b = 6$ mm. Assuming that the space between the conductors is filled with a nonmagnetic material, calculate the inductance per unit length.

8.44 Show that the mutual inductance between the rectangular loop and the infinite line current of Figure 8.4 is

$$M_{12} = \frac{\mu b}{2\pi} \ln\left[\frac{a + \rho_o}{\rho_o}\right]$$

Calculate M_{12} when $a = b = \rho_o = 1$ m.

Section 8.9—Magnetic Energy

8.45 A coaxial cable consists of an inner conductor of radius 1.2 cm and an outer conductor of radius 1.8 cm. The two conductors are separated by an insulating medium ($\mu = 4\mu_o$). If the cable is 3 m long and carries 25 mA current, calculate the energy stored in the medium.

8.46 In a certain region for which $\chi_m = 19$,

$$\mathbf{H} = 5x^2yz\mathbf{a}_x + 10xy^2z\mathbf{a}_y - 15xyz^2\mathbf{a}_z \text{ A/m}$$

How much energy is stored in $0 < x < 1, 0 < y < 2, -1 < z < 2$?

8.47 In a certain medium with $\mu = 4.5\,\mu_o$, $\mathbf{H} = 200\mathbf{a}_x + 500\mathbf{a}_y$ mA/m. Calculate the total energy stored in a $2 \times 2 \times 2$ cm cubical region centered at the origin.